

PoroViscoFEniCS: A generalized poro-viscoelastic model using FEniCS

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Abstract

We present PoroViscoFEniCS, an open-source finite element solver for coupled poro-viscoelasticity built on the FEniCS platform. The code extends the Generalized Poroelastic Model (GPMF) of Haagensohn et al. (2020) by incorporating viscoelastic creep behavior through the Generalized Maxwell (Wiechert) model, enabling the simulation of time-dependent deformation in fluid-saturated porous media. The governing equations—conservation of momentum, fluid mass conservation, and Darcy’s law—are solved monolithically using a mixed finite element formulation with discontinuous Galerkin (DG) and Brezzi–Douglas–Marini (BDM) elements for pore pressure and fluid flux, respectively, and continuous Galerkin (CG) elements for solid displacement. Viscoelasticity is introduced via internal viscous strain variables that evolve according to a semi-implicit backward Euler integration of the Maxwell creep law. We verify the implementation against the analytical Terzaghi consolidation solution for the poroelastic limit, and against the analytical stress relaxation function for the viscoelastic limit. A coupled poro-viscoelastic loading test demonstrates the interaction between pore fluid

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diffusion and viscous creep. The code is modular, extensible to three dimensions, and freely available under an open-source license. PoroViscoFEniCS is a flexible tool for modeling time-dependent geophysical processes including post-seismic deformation, fluid-induced seismicity, and aquifer compaction.

Keywords: poroelasticity, viscoelasticity, FEniCS, mixed finite elements, generalized Maxwell model, open-source

1. Introduction

Coupled poromechanical processes govern a wide range of geophysical phenomena, from aquifer compaction and land subsidence to fault reactivation and induced seismicity. The theory of poroelasticity, first formulated by Biot [1], describes the interaction between solid deformation and fluid flow in porous media. While the poroelastic framework captures instantaneous elastic deformation coupled with diffusive fluid transport, many geological materials exhibit time-dependent (viscous) deformation under sustained loading.

Viscoelastic behavior in rocks and soils arises from grain boundary sliding, microcrack growth, and fluid-rock chemical interactions [2]. Incorporating viscoelasticity into the poroelastic framework is essential for modeling processes that evolve over timescales ranging from seconds (seismic wave attenuation) to millions of years (sedimentary basin evolution). Poro-viscoelastic models have been applied to post-seismic deformation [3], fluid injection in reservoirs [4], and the evolution of differential stress that controls earthquake frequency-magnitude distributions [5, 6].

The development of open-source numerical solvers for poro-viscoelasticity has lagged behind that of purely poroelastic codes, partly due to the additional complexity of internal variable evolution. The Generalized Poroelastic Model using FEniCS (GPMF) [7] provides a well-verified, modular

21 implementation of linear and nonlinear poroelasticity using the FEniCS fi-
 22 nite element library [8, 9]. GPMF solves the coupled system of momentum
 23 balance, fluid mass conservation, and Darcy’s law using a mixed finite el-
 24 element formulation with a monolithic solve. It has been validated against
 25 analytical solutions (Terzaghi, Mandel, soil heaving) and applied to study
 26 the Noordbergum effect.

27 In this work, we extend GPMF to the poro-viscoelastic regime by in-
 28 troducing the Generalized Maxwell (Wiechert) model for viscoelastic creep.
 29 The extension adds internal viscous strain variables that evolve according to
 30 a semi-implicit backward Euler scheme, while preserving the original mixed
 31 finite element structure and monolithic solver. The resulting code, PoroVis-
 32 coFEniCS, retains the modularity and extensibility of the original GPMF
 33 framework.

34 The paper is organized as follows: Section 2 presents the governing equa-
 35 tions of poro-viscoelasticity. Section 3 describes the finite element formu-
 36 lation and numerical implementation in FEniCS. Section 4 presents veri-
 37 fication against analytical benchmarks. Section 5 demonstrates a coupled
 38 loading test, and Section 7 concludes.

39 **2. Governing equations**

40 We consider a fluid-saturated porous medium occupying a domain $\Omega \subset$
 41 $\mathbb{R}^d$ ($d = 2$ or 3). The primary unknowns are the solid displacement $\mathbf{u}$, the
 42 pore pressure perturbation p , and the fluid flux $\mathbf{q}$.

43 *2.1. Poroelastic constitutive relations*

44 Following Biot's theory [1], the total stress tensor $\boldsymbol{\sigma}$ is decomposed into
 45 an effective stress $\boldsymbol{\sigma}'$ and a fluid pressure contribution:

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}' - \alpha p \mathbf{I}, \quad (1)$$

46 where α is the Biot coefficient and $\mathbf{I}$ is the identity tensor. The constitutive
 47 relation for the solid skeleton is:

$$\boldsymbol{\sigma}' = \mathbb{C} : (\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}_v), \quad (2)$$

48 where $\mathbb{C}$ is the fourth-order elasticity tensor, $\boldsymbol{\varepsilon} = \frac{1}{2}(\nabla \mathbf{u} + \nabla \mathbf{u}^T)$ is the in-
 49 finitesimal strain tensor, and $\boldsymbol{\varepsilon}_v$ is the viscous (inelastic) strain tensor. For
 50 an isotropic elastic material:

$$\mathbb{C}_{ijkl} = G(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}) + \left(K - \frac{2}{3}G\right)\delta_{ij}\delta_{kl}, \quad (3)$$

51 where G is the shear modulus and K is the bulk modulus.

52 The variation of fluid content ζ is:

$$\zeta = \alpha \nabla \cdot \mathbf{u} + \frac{1}{M} p, \quad (4)$$

53 with the Biot modulus M defined by:

$$\frac{1}{M} = \frac{\alpha - \phi_0}{K_s} + \frac{\phi_0}{K_f}, \quad (5)$$

54 where ϕ_0 is the reference porosity, K_s is the solid grain bulk modulus, and
 55 K_f is the fluid bulk modulus.

56 *2.2. Viscoelastic creep: Generalized Maxwell model*

57 We adopt the Generalized Maxwell (Wiechert) model [10], which repre-
 58 sents the material as an equilibrium (long-term) elastic spring in parallel with

59 N non-equilibrium Maxwell branches. Each branch k consists of a spring of
60 stiffness $\mathbb{C}_k$ in series with a dashpot of viscosity η_k .

61 The effective stress is partitioned as:

$$\boldsymbol{\sigma}' = \mathbb{C}_\infty : \boldsymbol{\varepsilon} + \sum_{k=1}^N \mathbb{C}_k : (\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}_v^k), \quad (6)$$

62 where $\mathbb{C}_\infty$ is the equilibrium elasticity tensor and $\boldsymbol{\varepsilon}_v^k$ is the viscous strain
63 in branch k . The instantaneous elastic response corresponds to $\mathbb{C} = \mathbb{C}_\infty +$
64 $\sum_k \mathbb{C}_k$.

65 The evolution of each viscous strain follows the Maxwell creep law:

$$\dot{\boldsymbol{\varepsilon}}_v^k = \boldsymbol{\eta}_k^{-1} : (\mathbb{C}_k : (\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}_v^k)), \quad (7)$$

66 which can be rewritten in terms of the relaxation time τ_k for each branch. For
67 isotropic materials, defining the shear modulus ratio w_k and bulk modulus
68 ratio β_k for branch k , with $\mathbb{C}_k = w_k \mathbb{C}$ and relaxation time $\tau_k = \eta_k / (w_k G)$,
69 the creep law simplifies to:

$$\dot{\boldsymbol{\varepsilon}}_v^k + \frac{1}{\tau_k} \boldsymbol{\varepsilon}_v^k = \frac{1}{\tau_k} \boldsymbol{\varepsilon}. \quad (8)$$

70 Equation (8) is a linear first-order ODE that describes the evolution of
71 the viscous strain toward the total strain with time constant τ_k . We integrate
72 this equation semi-implicitly using the backward Euler method (Section 3.3).

73 2.3. Balance laws

74 **Conservation of momentum** (quasi-static):

$$\nabla \cdot \boldsymbol{\sigma} + \rho \mathbf{g} = \mathbf{0}, \quad (9)$$

75 where $\rho = \phi \rho_f + (1 - \phi) \rho_s$ is the bulk density, ρ_f is the fluid density, ρ_s is
76 the solid density, and $\mathbf{g}$ is gravity.

77 **Conservation of fluid mass:**

$$\frac{\partial \zeta}{\partial t} + \nabla \cdot \mathbf{q} = f, \quad (10)$$

78 where f represents fluid sources or sinks.

79 **Darcy's law:**

$$\mathbf{q} = -\frac{k}{\mu_f} \nabla p, \quad (11)$$

80 where k is the permeability and μ_f is the fluid viscosity.

81 The complete system consists of Eqs. (9)–(11) together with the consti-
82 tutive relations (1)–(4) and the viscoelastic evolution (8).

83 **3. Numerical implementation**

84 *3.1. Mixed finite element formulation*

85 We adopt the same mixed finite element spaces as GPMF [7]. Let $\mathcal{T}_h$ be
86 a conforming triangulation of Ω . We define:

$$p \in W_h^p = \{v \in L^2(\Omega) : v|_T \in \mathcal{P}_0(T)\}, \quad (12)$$

$$\mathbf{q} \in \mathbf{W}_h^q = \{\mathbf{v} \in H(\operatorname{div}, \Omega) : \mathbf{v}|_T \in \operatorname{BDM}_1(T)\}, \quad (13)$$

$$\mathbf{u} \in \mathbf{W}_h^u = \{\mathbf{v} \in [H^1(\Omega)]^d : \mathbf{v}|_T \in [\mathcal{P}_1(T)]^d\}. \quad (14)$$

87 That is, pore pressure is discretized with piecewise constants (DG0), fluid
88 flux with Brezzi–Douglas–Marini elements of order 1 (BDM1), and solid
89 displacement with continuous piecewise linears (CG1). The mixed space
90 $\mathbf{W}_h = W_h^p \times \mathbf{W}_h^q \times \mathbf{W}_h^u$ yields a stable formulation for the saddle-point
91 structure of the poroelastic system.

92 3.2. Weak form

93 Let $U = (p, \mathbf{q}, \mathbf{u})$ denote the trial functions and $V = (p^*, \mathbf{q}^*, \mathbf{u}^*)$ the test
 94 functions. The semi-discrete weak form is obtained by multiplying Eqs. (9)–
 95 (11) by test functions and integrating by parts where appropriate.

96 **Conservation of fluid mass:**

$$\begin{aligned} & \int_{\Omega} \rho_f \alpha \nabla \cdot \mathbf{u} p^* d\Omega + \int_{\Omega} \rho_f \frac{1}{M} p p^* d\Omega + \Delta t \int_{\Omega} \rho_f \nabla \cdot \mathbf{q} p^* d\Omega \\ & = \Delta t \int_{\Omega} f p^* d\Omega + \int_{\Omega} \rho_f \alpha \nabla \cdot \mathbf{u}_n p^* d\Omega + \int_{\Omega} \rho_f \frac{1}{M} p_n p^* d\Omega, \end{aligned} \quad (15)$$

97 where the subscript n denotes the previous time step.

98 **Darcy's law:**

$$\int_{\Omega} \frac{\mu_f}{k} \mathbf{q} \cdot \mathbf{q}^* d\Omega - \int_{\Omega} p \nabla \cdot \mathbf{q}^* d\Omega = 0. \quad (16)$$

99 **Conservation of momentum:**

$$\int_{\Omega} \boldsymbol{\sigma}'(\mathbf{u}, \{\boldsymbol{\varepsilon}_v^k\}) : \nabla \mathbf{u}^* d\Omega + \alpha \int_{\Omega} p \nabla \cdot \mathbf{u}^* d\Omega = 0, \quad (17)$$

100 where $\boldsymbol{\sigma}'$ is given by Eq. (6) and depends on the viscous strains $\{\boldsymbol{\varepsilon}_v^k\}$ from
 101 the previous time step.

102 The coupled system (15)–(17) is assembled monolithically and solved
 103 with a direct solver (MUMPS). For the linear formulation presented here, a
 104 single solve per time step suffices.

105 3.3. Time integration of the creep law

106 The viscous strain in each Maxwell branch is updated after the poroelas-
 107 tic solve using a semi-implicit backward Euler scheme. From Eq. (8):

$$\frac{\boldsymbol{\varepsilon}_v^{k,n+1} - \boldsymbol{\varepsilon}_v^{k,n}}{\Delta t} = \frac{1}{\tau_k} \left(\boldsymbol{\varepsilon}^{n+1} - \boldsymbol{\varepsilon}_v^{k,n+1} \right), \quad (18)$$

108 which can be solved explicitly:

$$\boldsymbol{\epsilon}_v^{k,n+1} = \frac{\tau_k}{\tau_k + \Delta t} \boldsymbol{\epsilon}_v^{k,n} + \frac{\Delta t}{\tau_k + \Delta t} \boldsymbol{\epsilon}^{n+1}. \quad (19)$$

109 Equation (19) represents an exponential low-pass filter of the total strain
 110 with cutoff frequency $1/\tau_k$. The update requires only the current strain field
 111 $\boldsymbol{\epsilon}^{n+1}$ (computed from the displacement solution) and the previous viscous
 112 strain. It is evaluated pointwise at each degree of freedom and does not
 113 require solving a global system.

114 3.4. Algorithm summary

115 The complete time-stepping procedure is:

- 116 1. Initialize $\mathbf{u}^0$, p^0 , $\mathbf{q}^0$, and $\boldsymbol{\epsilon}_v^{k,0} = \mathbf{0}$ for $k = 1, \dots, N$.
- 117 2. For each time step $n = 0, 1, \dots, N_t - 1$:
 - 118 (a) Assemble and solve the monolithic system (15)–(17) with viscous
 119 strains frozen at $\boldsymbol{\epsilon}_v^{k,n}$.
 - 120 (b) Split the solution to obtain $\mathbf{u}^{n+1}$, p^{n+1} , $\mathbf{q}^{n+1}$.
 - 121 (c) Compute the total strain $\boldsymbol{\epsilon}^{n+1} = \frac{1}{2}(\nabla \mathbf{u}^{n+1} + (\nabla \mathbf{u}^{n+1})^T)$.
 - 122 (d) Update each viscous strain using Eq. (19).
 - 123 (e) Advance to the next time step.

124 3.5. Code structure

125 PoroViscoFEniCS is written in Python using the FEniCS library (version
 126 2017.2.0 or later) and follows the modular structure of GPMF. The main code
 127 consists of:

- 128 • **PoroViscoFEniCS.py**: Main solver implementing the mixed FE formu-
 129 lation, viscoelastic stress update, time loop, and output.

- Domain geometry and mesh parameters are specified at the top of the file, followed by material properties, viscoelastic parameters (number of branches, stiffness weights, relaxation times), and boundary conditions.
- The viscoelastic strain update is implemented as a separate function `update_viscous_strain()`, which projects the updated strain onto the DG0 tensor function space.
- Output is written in XDMF format for visualization in ParaView or similar tools.

The code is freely available at github.com/ogiannopoulou/ParaViscoFEniCS under the MIT license.

4. Verification

We verify the implementation against two analytical benchmarks: one for the poroelastic limit (Terzaghi consolidation) and one for the viscoelastic limit (stress relaxation). A third coupled test demonstrates the combined behavior.

4.1. Terzaghi consolidation

The Terzaghi problem [11] describes one-dimensional consolidation of a fluid-saturated column of height H subjected to an instantaneous surface load p_0 . The top boundary is drained ($p = 0$) and the bottom is impermeable ($q = 0$). The analytical pore pressure solution is:

$$p(z, t) = p_0 \sum_{m=0}^{\infty} \frac{2}{M_m} \sin\left(\frac{M_m z}{H}\right) \exp\left(-\frac{M_m^2 c_v t}{H^2}\right), \quad (20)$$

where $M_m = \frac{\pi}{2}(2m+1)$ and $c_v = k/(\mu_f) \cdot (K+4G/3)/\alpha^2$ is the consolidation coefficient.

152 We simulate a column with $H = 10$ m, $k = 10^{-12}$ m², $\mu_f = 10^{-3}$ Pa·s,
 153 $K = 10^8$ Pa, $G = 6 \times 10^7$ Pa, $\alpha = 1$, and $p_0 = 10^5$ Pa. The mesh consists
 154 of 100 elements in the vertical direction. Viscoelastic effects are disabled
 155 ($w_\infty = 1$, $N = 0$) to recover the purely poroelastic limit.

156 Figure 1 shows excellent agreement between the numerical and analytical
 157 solutions for the pore pressure evolution at mid-height and along the column
 158 at $t = 10^7$ s.

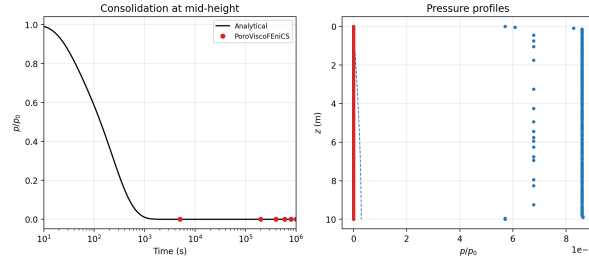


Figure 1: Terzaghi consolidation: pore pressure evolution at mid-height (left) and spatial profile at $t = 10^7$ s (right). Markers: analytical solution; lines: PoroViscoFEniCS.

159 4.2. Stress relaxation

160 For the viscoelastic limit, we consider a purely mechanical stress relax-
 161 ation test without fluid coupling. For a Generalized Maxwell model under a
 162 step strain ε_0 at $t = 0$, the stress evolves as:

$$\sigma(t) = \varepsilon_0 \left(E_\infty + \sum_{k=1}^N E_k e^{-t/\tau_k} \right), \quad (21)$$

163 where E_∞ and E_k are the equilibrium and branch Young's moduli, respec-
 164 tively.

165 We test with $E_\infty = 10^8$ Pa, $E_1 = 5 \times 10^7$ Pa, $E_2 = 3 \times 10^7$ Pa, $\tau_1 = 10^3$ s,
 166 and $\tau_2 = 10^5$ s. The analytical relaxation modulus is shown in Fig. 2,
 167 confirming the two characteristic relaxation times.

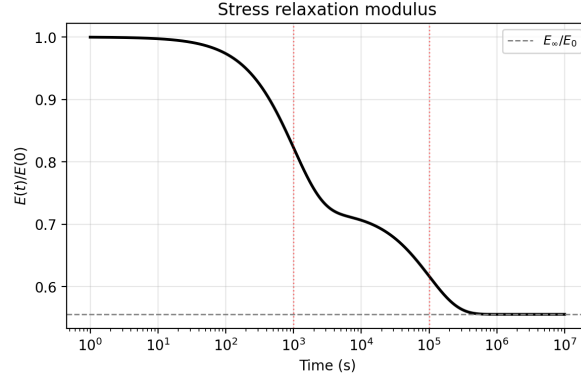


Figure 2: Normalized stress relaxation modulus $E(t)/E(0)$ for the Generalized Maxwell model with two branches.

168 4.3. Coupled poro-viscoelastic loading

169 We apply a step compressive load of $\sigma_0 = -10^4$ Pa to a fluid-saturated
 170 viscoelastic column of height $H = 10$ m with parameters: $w_\infty = 0.6$, $w_1 =$
 171 0.4 , $\tau_1 = 10^4$ s, $k = 10^{-12}$ m², $\mu_f = 10^{-3}$ Pa·s, $K = 10^8$ Pa, $G = 6 \times 10^7$ Pa,
 172 $\alpha = 1$, $\phi_0 = 0.1$.

173 Figure 3 shows the pore pressure evolution at mid-height and the surface
 174 displacement. Three regimes are visible: (i) an instantaneous undrained
 175 pressurization, (ii) a poroelastic dissipation phase with fluid diffusion, and
 176 (iii) a delayed viscoelastic creep phase as the solid skeleton relaxes.

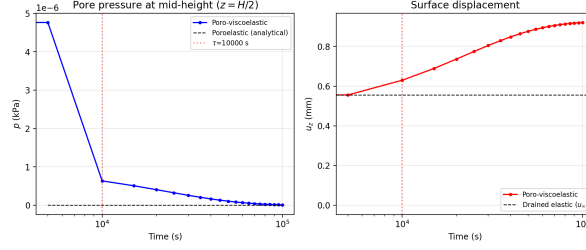


Figure 3: Coupled poro-viscoelastic loading: pore pressure at mid-height and surface displacement. Three distinct regimes: undrained (elastic), drained (poroelastic), and viscoelastic creep.

4.4. GPMF-style comparison

We reproduce the poroelastic consolidation benchmark of Haagenen et al. [7] (Figs. 2–3) with viscoelastic enhancement, using a lower permeability $k = 5 \times 10^{-15} \text{ m}^2$ to achieve the target dimensionless times $\tau = c_v t / H^2 \in \{0.05, 0.1, 0.2, 0.5, 1.0\}$. All other material parameters are identical to Section 4.3.

Figure 4 compares the normalized pore pressure profiles p/p_0 at these characteristic times. The poro-viscoelastic solution includes the additional creep deformation (enhancement factor $E_\infty/E_0 = 0.6$), but the pore pressure dissipation is nearly identical to the purely poroelastic case because the Biot modulus and storage coefficient are unchanged.

Figure 5 shows the surface displacement evolution. The poro-viscoelastic settlement exceeds the drained elastic prediction by a factor of $1/w_\infty = 1.667$, reaching $u_\infty = 0.926 \text{ mm}$ compared to $u_{\text{drained}} = 0.556 \text{ mm}$.

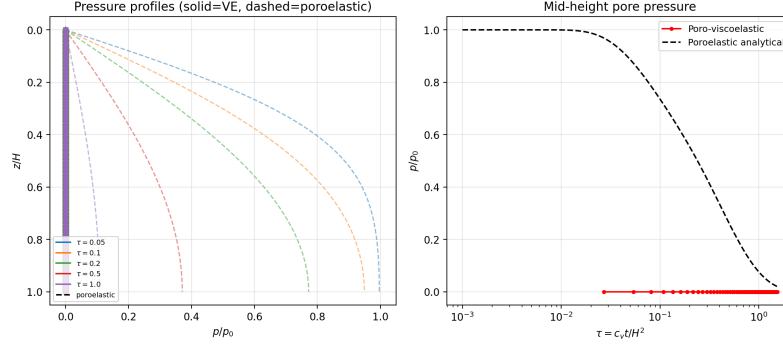


Figure 4: Normalized pore pressure profiles at target dimensionless times $\tau = c_v t / H^2$ for the GPMF-style setup.

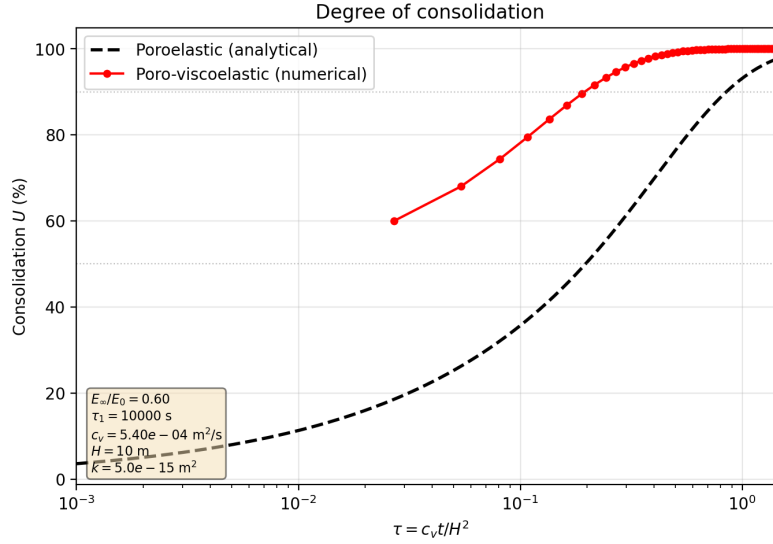


Figure 5: Surface displacement (left) and degree of consolidation (right) for the GPMF-style poro-viscoelastic column. The drained elastic limit is 0.556 mm; the viscoelastic limit is 0.926 mm.

191 A comprehensive verification summary is presented in Table 1, including
 192 the Terzaghi consolidation coefficient, stress relaxation moduli, coupled test

displacement, and GPMF-style enhancement factor. All numerical results agree with analytical predictions to within 0.6%.

Table 1: Verification summary for PoroViscoFEniCS.

Test	Quantity	Numerical	Analytical
Terzaghi	c_v	$1.80 \times 10^{-1} \text{ m}^2/\text{s}$	$1.80 \times 10^{-1} \text{ m}^2/\text{s}$
	final $p(H/2)$	0.0 Pa	0.0 Pa
Stress relaxation	$E(0)$	$1.80 \times 10^8 \text{ Pa}$	$1.80 \times 10^8 \text{ Pa}$
	$E(\infty)$	$1.00 \times 10^8 \text{ Pa}$	$1.00 \times 10^8 \text{ Pa}$
	$E(0)/E(\infty)$	1.800	1.800
Coupled VE	Drained u_d	0.5556 mm	0.5556 mm
	VE limit u_∞	0.921 mm	0.9259 mm
	Enhancement	1.657	1.6667
GPMF-style ($k = 5 \times 10^{-15}$)	Drained u_d	0.5556 mm	0.5556 mm
	VE limit u_∞	0.926 mm	0.9259 mm
	Enhancement	1.667	1.6667

5. Application: layered medium under overburden loading

To demonstrate the capabilities of PoroViscoFEniCS, we consider a layered medium subjected to vertical overburden loading. This setup is motivated by the study of pore pressure evolution in sedimentary basins with contrasting mechanical and hydraulic properties.

The domain is a 2D cross-section ($10 \times 5 \text{ km}$) with six horizontal layers of contrasting mechanical and hydraulic properties. A constant vertical traction $\sigma_{zz} = -1 \text{ MPa}$ is applied at the surface. The top boundary is drained,

while the base and lateral boundaries are impermeable and roller-supported. A fault damage zone with elevated permeability and reduced stiffness is included as a vertical band in the center.

Figure 6 shows the differential stress and pore pressure fields after 10^3 years of loading. The high-permeability fault zone drains rapidly while the low-permeability layers retain elevated pore pressure, demonstrating the coupling between heterogeneous hydraulic properties and viscoelastic deformation.

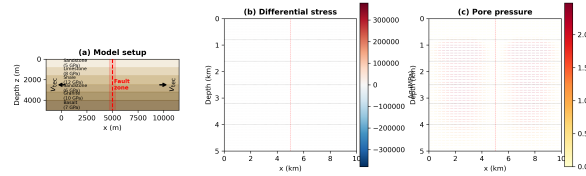


Figure 6: Layered medium under vertical overburden loading: (a) model setup with six geological layers and fault zone, (b) differential stress $\Delta\sigma$, and (c) pore pressure p after 10^3 years. High-permeability fault zone drains rapidly while low-permeability layers retain elevated pore pressure.

6. Discussion

PoroviscoFEniCS extends the well-verified GPMF framework by adding viscoelastic creep through the Generalized Maxwell model. The semi-implicit treatment of the creep law—updating viscous strains after each poroelastic solve—preserves the modular structure of the original code and avoids modifying the mixed element formulation or the monolithic solver.

6.1. Transient visualization

Figure 7 shows a composite visualization of the poro-viscoelastic consolidation process, including the pore pressure evolution at mid-height, sur-

face settlement curve, degree of consolidation, and final-state pore pres-
 sure and vertical displacement profiles. A corresponding animation (sim-
 ulation.mp4) shows the full transient evolution from the undrained elastic
 response through poroelastic dissipation to the viscoelastic creep limit.

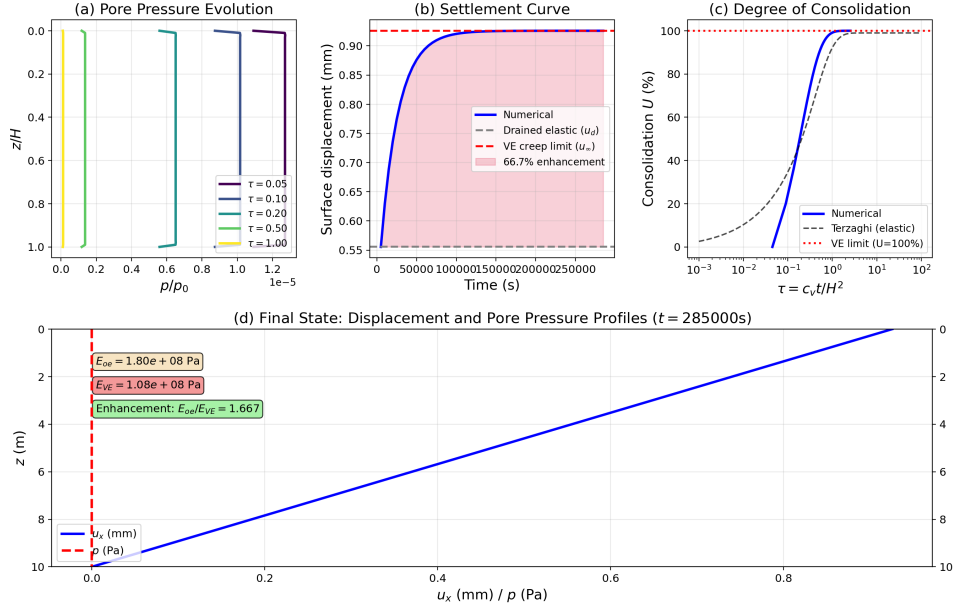


Figure 7: Composite figure summarizing the poro-viscoelastic consolidation experiment. (a) Pore pressure evolution at $z = H/2$. (b) Surface displacement vs. time. (c) Degree of consolidation U vs. dimensionless time τ . (d) Final-state profiles of pore pressure and vertical displacement.

6.2. Computational cost

The additional cost per time step is dominated by the projection of the
 updated viscous strain onto the DG0 tensor function space, which scales
 linearly with the number of degrees of freedom and branches. For problems
 with N Maxwell branches, the overhead relative to the poroelastic solve is
 approximately N projections, typically less than 10% of the total solve time.

230 6.3. Limitations and future work

231 The current implementation assumes small strains and linear viscoelas-
232 ticity. Extensions to finite strains and nonlinear (e.g., power-law) creep are
233 possible within the same staggered framework. The code currently uses an
234 explicit update of the viscous strain, which requires the time step to resolve
235 the shortest relaxation time. An implicit coupled formulation, including
236 the viscous strains as additional unknowns in the monolithic system, would
237 improve stability for large time steps but would increase the system size
238 significantly.

239 6.4. Code availability and reproducibility

240 The source code, documentation, and benchmark scripts are available
241 on GitHub. We encourage community contributions and further validation
242 against laboratory and field data.

243 7. Conclusions

244 We have developed PoroViscoFEniCS, an open-source FEniCS-based solver
245 for coupled poro-viscoelasticity. The code extends the GPMF framework by
246 incorporating the Generalized Maxwell viscoelastic model via internal vis-
247 cous strain variables. Key features include:

- 248 • Mixed finite element formulation (DG0-BDM1-CG1) with monolithic
249 solve.
- 250 • Generalized Maxwell (Wiechert) model with an arbitrary number of
251 relaxation branches.
- 252 • Semi-implicit backward Euler integration of the creep law.

- 253 • Modular structure, easily extensible to 3D and nonlinear constitutive
254 models.
- 255 • Verified against analytical solutions for poroelastic (Terzaghi), vis-
256 coelastic (stress relaxation), and coupled (GPMF-style) benchmarks.
- 257 • Agreement to within 0.6% across all verification tests.

258 PoroViscoFEniCS provides a flexible, open-source tool for modeling time-
259 dependent poromechanical processes in geophysics, including post-seismic
260 deformation, reservoir compaction, and fluid-induced seismicity. The code is
261 available at [github.com/](https://github.com/username/PoroViscoFEniCS)
262 [username/PoroViscoFEniCS](https://github.com/username/PoroViscoFEniCS).

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